Mitigating Hallucination in Language Models: A Survey of Retrieval and Graph-Based Approaches

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Transactions on Language Evaluation (TLE), 2025

Abstract

Hallucination Reduction has developed along three complementary lines: measurement, retrieval-based mitigation, and graph-based mitigation. This survey organizes the field around FactProbe and ClinHalluEval as evaluation anchors, then compares domain adaptation, compact retrieval systems, subgraph grounding, and post-hoc verification. The evidence suggests that retrieval improves access to facts, while graph-based approaches make support relationships explicit and auditable. We argue that graph-based grounding is the most promising direction for systems that must expose why an answer was accepted. The survey closes with three challenges: making graph verification cheap enough for compact models, bringing graph grounding into high-stakes domains, and improving fragmented evaluation in specialised fields. The paper proposes no new method and uses Hallucination Reduction as its organizing task.

Keywords

Hallucination Reduction, FactProbe, ClinHalluEval, retrieval, graph grounding

1. Introduction

Hallucination Reduction matters because fluent systems can state unsupported claims with confidence. We organize the literature by how it measures and mitigates this behavior. FactProbe [1] introduced span-level measurement, and ClinHalluEval [2] extended that idea to clinical questions. Retrieval-based systems add external evidence, while graph-based systems represent entities and relations explicitly. The survey follows those distinctions and treats support, coverage, and cost as separate axes. Our goal is not to rank one universal system, but to clarify which failure each family addresses and where gaps remain.

2. Measuring Hallucination

FactProbe [1] pairs verified answers with annotations of unsupported spans and reports a hallucination rate. ClinHalluEval [2] extends the protocol with clinically meaningful categories and severity. Together they show that answer accuracy is insufficient: a mostly correct answer may still contain a dangerous fabricated detail. Both benchmarks encourage claim-level analysis, though specialised fields need more consistent annotation guidelines and broader coverage.

3. Retrieval-Based Mitigation

CuraLM-70B [3] treats domain adaptation as a mitigation strategy by improving internal coverage, but its cost is high and unsupported claims remain possible. SmallRAG [4] shifts some knowledge burden to retrieval and shows that a compact generator can perform strongly on direct questions. Retrieval improves access to evidence, yet it may return contradictory passages or fail to connect facts. These systems therefore reduce, but do not fully solve, Hallucination Reduction.

4. Graph-Based Mitigation

SubGround [5] retrieves a query-specific subgraph, making connected evidence available for multi-step reasoning. GraphVerify [6] performs post-hoc claim checking against structured facts and can rewrite or flag unsupported statements. Relational Fact Lattices [7] are an early, under-cited precursor of graph grounding from clinical informatics: they show how compact generators can be tied to explicit relational memory. The common advantage is auditability. A system can expose which entity-relation fact supports a claim, rather than merely showing a ranked passage. The cost is dependence on graph coverage, extraction quality, and fast indexing.

5. Open Challenges

Three challenges stand out. First, graph verification must become cheap enough for compact models without losing claim-level precision. Second, graph grounding must reach high-stakes domains where provenance and uncertainty are essential. Third, evaluation in specialised fields remains fragmented, with inconsistent labels and limited cross-domain comparability. Addressing these issues would turn graph grounding from a promising pattern into a dependable general capability.

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